

CHE 435/525 *Process Systems Analysis and Control*

Lecture 13: **Feedback Control and Feedback Loop**

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Recapture: Frequency response analysis

- Frequency response: A.R. = $\hat{A}/A = |G(i\omega)|$, $\phi = \angle G(i\omega)$
- Bode diagram: A (log scale) v.s. ω (log scale), ϕ v.s. ω
- First-order system $G(s) = \frac{K}{\tau s + 1}$; Lead-lag system $G(s) = \frac{1 + \tau_0 s}{1 + \tau_1 s}$;
Non-minimum-phase element $G(s) = \frac{1 - \tau s}{1 + \tau s}$,
- Higher-order systems in a general form:

$$G(s) = K \frac{(1 + \bar{\tau}_1 s) \dots (1 + \bar{\tau}_m s)}{(1 + \tau_1 s) \dots (1 + \tau_n s)} e^{-\theta s} \quad (K > 0)$$

Use in Practice: Frequency-domain system identification

- An accurate approach to identify the t.f. model of an unknown system

Guidelines

1 Obtain the Bode diagram

- ▶ Choose many ω values, let $u(t) = A \sin \omega t$ and measure $y(t)$
- ▶ Determine A.R. = $\hat{A}/A$ and ϕ from $y(t) = \hat{A} \sin(\omega t + \phi)$

2 Check the asymptotic values/slopes on the Bode diagram

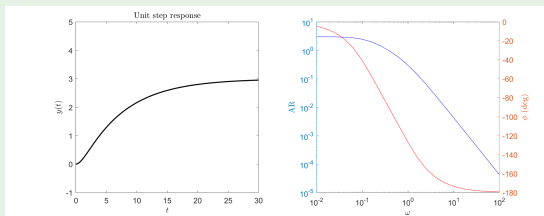
3 Choose factors to match the Bode diagram

- ▶ Lag: $\frac{1}{1+\tau s}$... AR  slope 0 to -1 , ϕ  value 0 to $-\pi/2$
- ▶ Lead: $1 + \tau s$... AR  slope 0 to $+1$, ϕ  value 0 to $+\pi/2$
- ▶ NMP: $\frac{1-\tau s}{1+\tau s}$... AR = 1, ϕ  value 0 to $-\pi$
- ▶ Delay: $e^{-\theta s}$... AR = 1, ϕ drops from 0 to $-\infty$
- ▶ Integrator: $\frac{1}{s}$... AR  slope -1 , $\phi = -\pi/2$
- ▶ Differentiator: s ... AR  slope $+1$, $\phi = +\pi/2$
- ▶ Underdamped 2nd-order: $\frac{K}{\tau^2 s^2 + 2\tau \zeta s + 1}$: AR  slope 0 to -2 , ϕ  value 0 to $-\pi$

Frequency-domain v.s. time-domain identification

Step-response data v.s. Frequency-response data

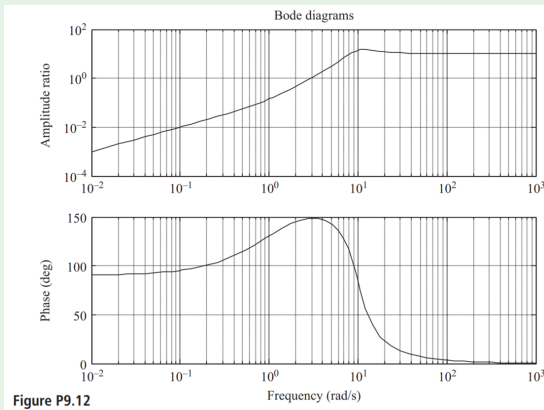
$$G(s) = \frac{3}{7s^2 + 8s + 1}$$



- Identification from step response
 - ▶ Likely a 2nd-order system, close to 1st-order, not sure if higher-order
- Identification from frequency responses
 - ▶ Two lags for sure!

An Example for frequency-domain system identification

** Frequency-domain system identification [Ex. 9.12, Kravaris]



• Observation and analysis

- ▶ A.R. slope: initially +1, then +2, then 0
- ▶ ϕ values: initially 90° , then rises, but countered by a decrease to 0°

- Observation and analysis

- ▶ A.R. slope: initially +1, then +2, then 0 — ∴ 3 factors exist

- ① first factor =  , with slope +1

- ② second factor =  , with slope 0 → +1

- ③ third factor =  , with slope 0 → -2

- ▶ ϕ values: initially 90° , then rises, but countered by a decrease to 0°

- ① first factor = s , 90°

- ② second factor = $1 + \tau s$, $0^\circ \rightarrow 90^\circ$

- ③ third factor = 2^{nd} -order lag, $0^\circ \rightarrow -180^\circ$

- The t.f. should be

$$G(s) = \boxed{\frac{Ks(1 + \hat{\tau}s)}{\tau^2 s^2 + 2\tau\zeta s + 1}}.$$

- Obtain the parameters – express A.R. and ϕ

$$|G(i\omega)| = \frac{K\omega\sqrt{1 + \omega^2\hat{\tau}^2}}{\sqrt{(1 - \omega^2\tau^2)^2 + 4\omega^2\tau^2\zeta^2}}$$

$$\angle G(i\omega) = \frac{\pi}{2} + \arctan \omega\hat{\tau} - \boxed{\angle(1 - \omega^2\tau^2 + 2i\omega\tau\zeta)}$$

- When $\omega \ll 1$: A.R. $\approx K\omega$

★ Check: $\omega = 10^{-1}$, A.R. $\approx 10^{-2} \therefore K \approx 1/10$

- When $1 \leq \omega \leq 10$: A.R. $\approx K\omega^2\hat{\tau}$

★ Check: $\omega = 3$, A.R. $\approx 1 \therefore \hat{\tau} \approx 10/9$

- When $\omega \gg 10$: A.R. $\approx K\hat{\tau}/\tau^2$

★ Check: $\omega = 1000$, A.R. $\approx 10 \therefore \tau \approx 1/\sqrt{90}$

- When $\omega \approx 10$, $\phi \approx \pi - \angle(1 - \omega^2\tau^2 + 2i\omega\tau\zeta)$

★ Check: $\phi = \pi/4$ corresponds to $\omega = 15$.

★ $\phi = \pi/4$ means $1 - \omega^2\tau^2 < 0$ and $2\omega\tau\zeta = \omega^2\tau^2 - 1 \dots \therefore \zeta = 3/2\sqrt{10}$

- DONE!

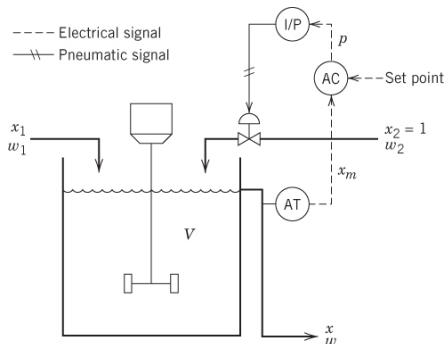
$$G(s) = \boxed{\frac{\frac{1}{9}s^2 + \frac{1}{10}s}{\frac{1}{90}s^2 + \frac{1}{10}s + 1}}$$

Learning objectives of this lecture

- 1 Explain schematically the role of **feedback controllers**
- 2 Recognize the time-domain and transfer-function forms of **PID controllers**
- 3 Represent the **closed-loop system** in a **block diagram**

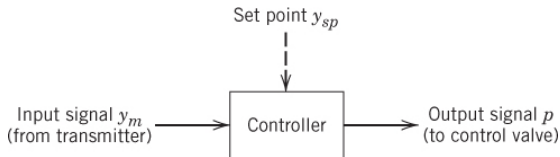
Recall: What is a feedback controller?

- A device to adjust **manipulated variables** in a compensatory way
- Typically, it uses a comparison of **controlled variables** to their setpoints



AT: analyzer transmitter (composition sensor), AC: controller
 p : electric current signal, I/P: current-to-pressure transducer

Block representation

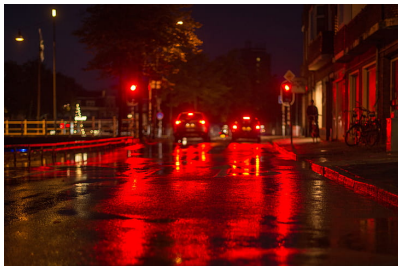


- Generally, the controller can be **any** algorithm, as long as the controller output p depends on the past y_m and y_{sp}
- Less generally, the controller can be designed as a **linear system**

$$P'(s) = G_1(s)Y_{sp}(s) + G_2Y_m(s)$$

- Even less generally, $P'(s) = G_c(s)E(s)$, where $E(s) = Y_{sp}(s) - Y_m(s)$ (i.e., $e(t) = y_{sp}(s) - y_m(s)$) is the **error signal**
 - ▶ Controller is a t.f. $G_c(s)$
- How to design the form of $G_c(s)$? (What kind of t.f. can it be?)

“Natural” design of controller: Three control modes



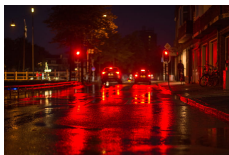
Controller output p : Force on the brake
Error e : Speed of the car & Distance to the stop line

① Proportional (P) control

$$p(t) = \bar{p} + K_c e(t)$$

- ▶ $\bar{p}$: steady state value, i.e., the value of $p(t)$ when $e(t) = 0$
- ▶ K_c : controller gain

“Natural” design of controller: Three control modes



Is your controller really that simple? ... Likely it contains more.

② Integral (I) mode

$$p(t) = K_I \int_0^t e(t^*) dt^*$$

- ▶ K_I : integration coefficient
- ▶ Proportional-integral controller (PI): $p(t) = \bar{p} + K_c e(t) + K_I \int_0^t e(t^*) dt^*$
- ▶ By writing $K_I = K_c / \tau_I$, we have

$$p(t) = \bar{p} + K_c \left[e(t) + \frac{1}{\tau_I} \int_0^t e(t^*) dt^* \right]$$

where τ_I is called **integral time (constant)**

Offset

- Why use I-mode?

- ▶ As long as the error remains, keep changing $p(t)$... until error totally vanishes
- ▶ This (hopefully) eliminates the error between y_m and y_{sp} ultimately
 - ★ **Offset** – steady-state value of $y(t) - y_{sp}(t)$, often in percentages
 - ★ I-mode achieves **offset-free control** (to be proved later)

“Natural” design of controller: Three control modes



What if there is a rapidly decelerating car ahead of you?

③ Derivative (D) mode

$$p(t) = K_D \frac{de(t)}{dt}$$

- ▶ Write $K_D = K_c \tau_D$, where τ_D is called **derivative time**
- ▶ Proportional-derivative controller (PD):

$$p(t) = \bar{p} + K_c \left[e(t) + \tau_D \frac{de(t)}{dt} \right]$$

or proportional-integral-derivative controller (PID):

$$p(t) = \bar{p} + K_c \left[e(t) + \frac{1}{\tau_I} \int_0^t e(t^*) dt^* + \tau_D \frac{de(t)}{dt} \right]$$

TF of PID controller

$$p(t) = \bar{p} + K_c \left[e(t) + \frac{1}{\tau_I} \int_0^t e(t^*) dt^* + \tau_D \frac{de(t)}{dt} \right]$$

$$\frac{P'(s)}{E(s)} = G_c(s) = K_c \left(1 + \frac{1}{\tau_I s} + \tau_D s \right)$$

- $G_c(s)$ is the TF of the controller.

Derivative filter

- Wait... There is no "process" that can appear like $K_D s$
 - ▶ For any physically realizable t.f., degree of denominator $\geq$ degree of numerator! (But $0 \geq 1$ does not hold.)
 - ▶ From a frequency-response point-of-view, such a controller would respond drastically to high-frequency noises.
- Filtered PID controller:
 - ▶ "Parallel with derivative filter":
$$G_c(s) = K_c \left[1 + \frac{1}{\tau_I s} + \frac{\tau_D s}{\alpha \tau_D s + 1} \right] \quad (\alpha \ll 1)$$
 - ▶ "Series with derivative filter": $G_c(s) = K_c \left[1 + \frac{1}{\tau_I s} \right] \frac{\tau_D s + 1}{\alpha \tau_D s + 1} \quad (\alpha \ll 1)$

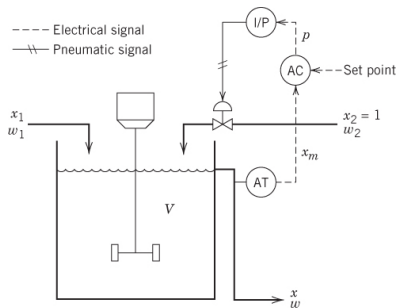
Be aware!

- After all, PID is just a specific form of controllers
 - ▶ PID is usually **not optimal**, and may even fail to work or work badly. There even exist processes that cannot be controlled with PID.
 - ▶ In general, controllers can be any t.f. or even any algorithm. There is no sound reason to restrict to PID unless you have to.
 - ▶ Engineers need to know the process models first, and then design a suitable controller (instead of being ignorant of the process and expecting a magical “tuning rule”)!

Towards closed-loop system analysis

- Now the central problem of process control is **how to design the controller**
- ... based on the **analysis** of how the controller design affects **regulation and tracking**
- ... i.e., the effect of y_{sp} on y , and the effect of d (disturbances) on y , **under control**
- So, we need to put together everything in the loop

Components of a feedback control system



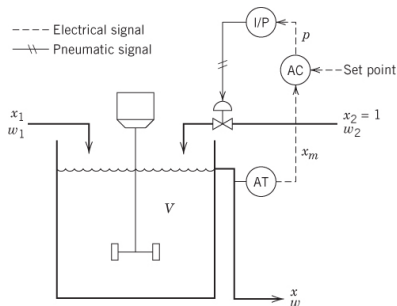
- 1 Process (plant): x (controlled) affected by w_2 (manipulated) and x_1 (disturbance)

$$X'(s) = \frac{K_1}{\tau_1 s + 1} X'_1(s) + \frac{K_2}{\tau_2 s + 1} W'_2(s)$$

- 2 Sensor: approximately a first-order system

$$X'_m(s)/X(s) = G(s) = \frac{K_m}{\tau_m s + 1} X(s)$$

Components of a feedback control system



- ③ Controller: a PI for example

$$\frac{P'(s)}{E(s)} = K_c \left[1 + \frac{1}{\tau_I s} \right], \quad E(s) = \tilde{Y}_{sp}(s) - Y_m(s) = K_m Y_{sp}(s) - Y_m(s)$$

- ④ Current-to-pressure transducer: $P'_t(s) = K_{IP} P'(s)$

- ⑤ Control valve: $W'_2(s)/P'_t(s) = K_v/(\tau_v s + 1)$

Block diagram of the feedback control system

